

Random Features for Discrete Amortized Samplers

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TL;DR

- discrete amortized samplers (e.g., GFlowNets) suffer from **underdetermination**, **limited expressiveness**, and **inefficient exploration**,
- Mixture Model Augmentation (MMA)** injects **continuous random features** into the policy, giving a **mixture of Markov processes** that **provably boosts expressivity**,
- across three benchmarks, MMA **accelerates convergence** and **improves the fit**.

I. BACKGROUND

We sample $x \sim \pi(x) = Z^{-1}R(x)$ over a discrete, **compositional** set $\mathcal{X}$ with $Z = \sum_x R(x)$ intractable; each x is the sink of a **state-graph** DAG built from a seed s_o . A **forward policy** p_F matches the reward marginal,

$$p_{\top}(x) = \sum_{\tau: s_o \rightsquigarrow x} p_F(s_o, \tau) \propto R(x), \quad (1)$$

tied to R through a **backward policy** p_B by the **trajectory-balance (TB)** loss,

$$\mathcal{L}_{TB} = \mathbb{E}_{\tau \sim p_E} \left(\log \frac{Z p_F(s_o, \tau)}{R(x) p_B(x, \tau)} \right)^2. \quad (2)$$

II. THREE CHALLENGES

C1. Underdetermination. Infinitely many (p_F, p_B) satisfy TB.

C2. Limited expressiveness. A neural policy may represent **no** solution; e.g., 1-WL GNNs alias non-isomorphic states.

C3. Inefficient exploration. A single policy induces **state aliasing**: near-identical embeddings for states that need different actions.

III. MIXTURE MODEL AUGMENTATION

Attach a **continuous random feature** γ —**constant along each trajectory**—to every state. This serves as an auxiliary variable, mirroring the momentum variable in HMC-like samplers.

Augmented state graph. Lift $\mathcal{G}$ to $\bar{\mathcal{G}} = (\mathcal{S} \times \Gamma, \mathcal{X} \times \Gamma)$ with $(s, \gamma) \rightarrow (s', \gamma')$ iff $s \rightarrow s'$ and $\gamma = \gamma'$.

Marginalizing γ out gives a **convex mixture of Markov processes** — distinct γ index distinct policies, resolving C1:

Mixture semantics. For $\gamma_o \sim p(\gamma_o)$,

$$p_{\top}(s_t) = \int_{\Gamma} p(\gamma_t) \bar{p}_{\top}(s_t | \gamma_t) d\gamma_t. \quad (3)$$

With $R(x, \gamma) = R(x) p(\gamma)$ (same Z), we target $\bar{p}_{\top}(x, \gamma) \propto R(x, \gamma)$ via the **augmented TB loss** on the encoder $\eta(s, \gamma) = \varphi(s) \oplus \psi(\gamma)$, so MMA plugs into **any** existing sampler unchanged.

IV. EXPRESSIVITY: RNI GNNs

Adding γ is **random node initialization (RNI)**: an RNI GNN is a GNN **ensemble** and a universal approximator.

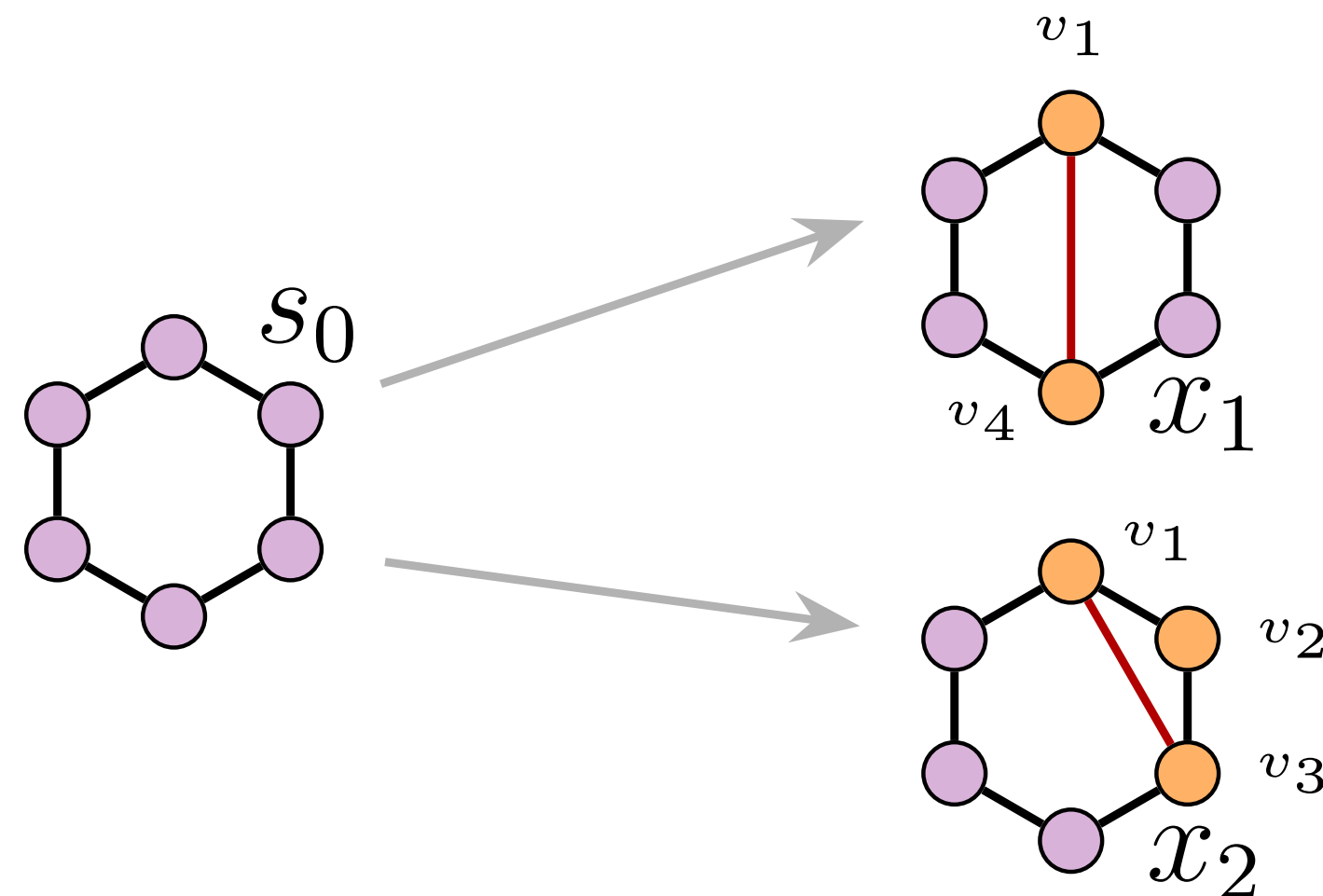


Figure 1: The 6-cycle s_o has non-isomorphic children x_1 (chord $v_1 -- v_4$) and x_2 (chord $v_1 -- v_3$).

A 1-WL p_F gives identical node embeddings, so it is **forced uniform** over x_1, x_2 and cannot fit $R_1 \neq R_2$. Random features $\bar{h} = h \oplus \gamma$ break the symmetry,

$$\bar{p}_F((s_o, \gamma), (x_j, \gamma)) \propto \exp\{\varphi(\bar{h}_{v_1} + \bar{h}_{v_j'})\}, \quad (4)$$

fitting **any** target and overcoming C2.

V. EXPERIMENTS

MMA **speeds up convergence** and **improves the fit** across Hypergrid, Lazy Random Walk, and Set Generation ($|\mathcal{X}| \sim 10^9$).

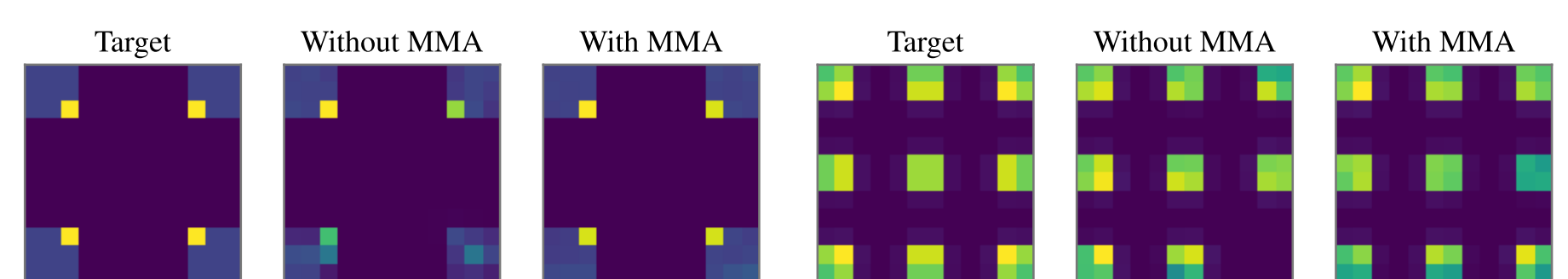


Figure 2: **Hypergrid** — CORNERS (left) and GAUSSIAN (right): MMA provides a better goodness-of-fit to the target.

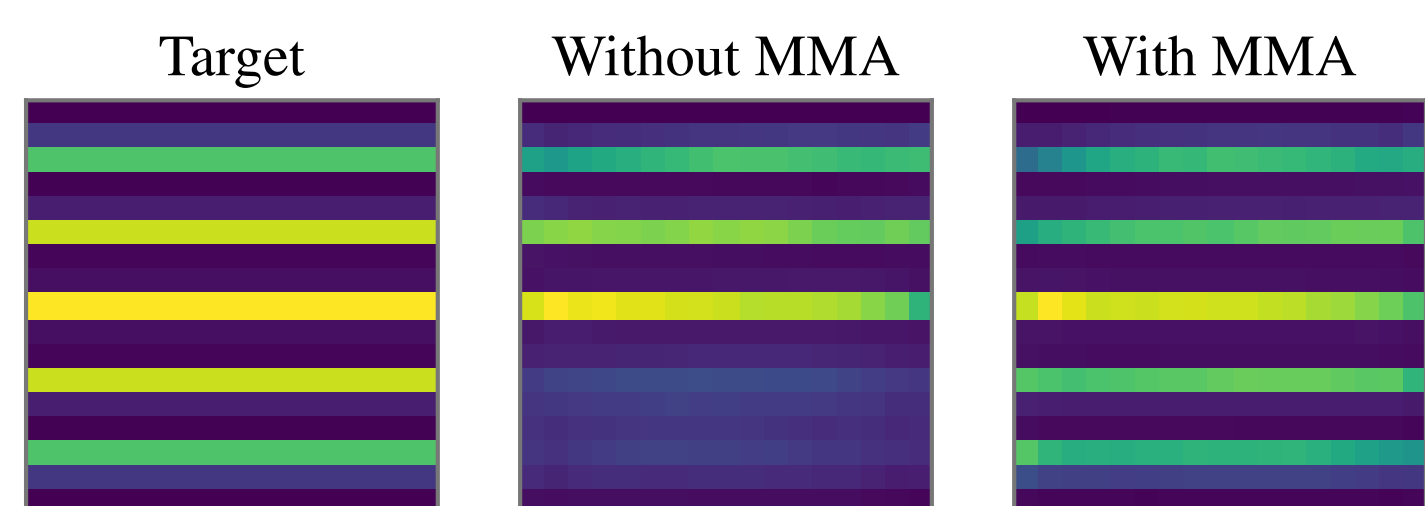


Figure 3: **Lazy Random Walk**: MMA recovers the striped target the un-augmented sampler washes out.

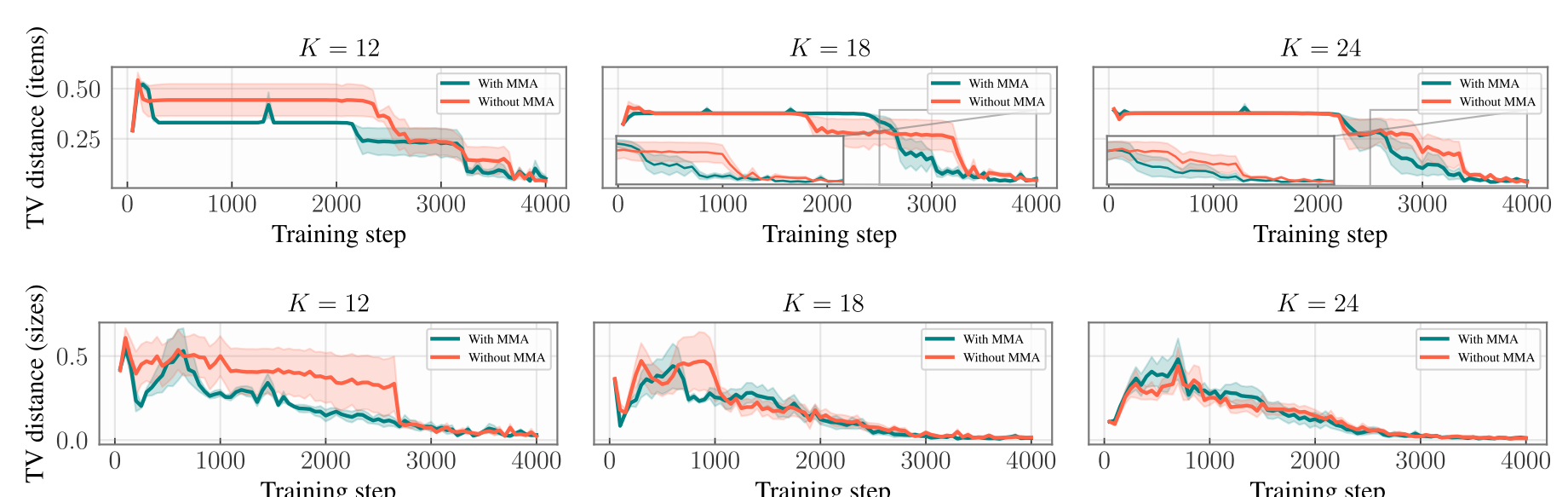


Figure 4: **Set Generation**: TV distance on item (top) and set-size (bottom) marginals vs. training step, $K \in \{12, 18, 24\}$. MMA converges faster on both.

Conclusions. MMA recasts a discrete sampler as a **mixture of Markovian samplers** via one auxiliary variable γ — **provably boosting expressivity (C2)** and **improving convergence (C1, C3)** at **no** extra trajectory length.